The complex Grothendieck constant

New bounds, machine-checked in Lean

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https://github.com/agalatan/grothendieck-lean

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$$1.401697117301 \;<\; K_G^{\mathbb{C}} \;<\; 1.404598149141$$

Both bounds are theorems of a Lean 4/Mathlib development, resting on explicitly stated analytic and computer-assisted inputs (Section 3).

Only the sharp upper target is active; its fourteen finite-band Gaussian estimates and energy-table accuracy remain to be proved.

Abstract

Grothendieck's inequality bounds how much a semidefinite (vector) relaxation of a bilinear optimisation problem can exceed the true optimum. The best constant in the complex case, $K_G^{\mathbb{C}}$, has been known since 1987 to lie roughly in $[1.338, 1.405]$; its first two decimals were settled only in September 2026. We improve both bounds, to an interval of width 0.0029, and verify the arguments in Lean. This note explains the problem, the two constructions, exactly what the computer checks, the new inequalities and obstructions found along the way, and why the remaining gap is hard: the two sides are dual optimisation problems, and the obstruction on both is the nonlinear (cubic and higher) structure of bounded functions of Gaussian vectors.

1 The question

Let $A = (a_{ij})$ be an $m \times n$ complex matrix. Compare

$$C(A) = \sup_{|\varepsilon_i|=|\delta_j|=1} \Big|\sum_{i,j} a_{ij}\,\overline{\varepsilon_i}\,\delta_j\Big|, \qquad V(A) = \sup_{\|x_i\|=\|y_j\|=1} \Big|\sum_{i,j} a_{ij}\,\langle x_i, y_j\rangle\Big|,$$

the first over unimodular scalars, the second over unit vectors in any complex Hilbert space. Clearly $C(A) \le V(A)$. Grothendieck proved in 1953 that $V(A) \le K\,C(A)$ for a constant K independent of m, n and the dimension. The smallest such K is the *complex Grothendieck constant* $K_G^{\mathbb{C}}$.

Why it matters.

- *Optimisation.* $V(A)$ is a semidefinite program; $C(A)$ is a hard combinatorial problem. $K_G^{\mathbb{C}}$ is the best approximation ratio of the semidefinite relaxation, and rounding algorithms for it are exactly upper-bound proofs.
- *Functional analysis.* K_G is the norm of the identity between the injective and projective tensor norms on $\ell_\infty \otimes \ell_\infty$, and controls absolutely summing operators (Grothendieck's original setting).

- *Quantum information.* The real constant $K_G^{\mathbb{R}}$ is the largest ratio between quantum and classical values of two-player XOR games (Tsirelson); the finite-order constants $K_G(d)$ govern the nonlocality of qubit states.

History.

year	who	lower bound	upper bound
1953	Grothendieck		$K < \infty$
1984	Davie	1.33807	
Sep 2026	Guo, Fang, Lu	1.35584631827	
Sep 2026	Heilman, Jones, Malavolta	1.4	archived
Oct 2026	**this work**	**1.401697117301**	**1.404598149141**

Haagerup's integral $1/\gamma^\dagger = 1.404575934\ldots$, $\gamma^\dagger = \int_0^{\pi/2} \cos^2 t\,(1+\sin^2 t)^{-1/2} dt$, has been proposed as the exact value. It lies inside our interval, $2.22 \cdot 10^{-5}$ below the upper bound. For the real constant the corresponding recent results are $1.773 \le K_G^{\mathbb{R}} \le 1.7799$ (Heilman–Jones–Malavolta, October 2026).

2 Two dual constructions

Both bounds use the same Gaussian picture. Let $Z = (z_1, \ldots, z_n)$ be a standard complex Gaussian vector and decompose $L^2(\mathbb{C}^n, \gamma)$ into the *Wick chaoses* $H_{p,q}$ (polynomials of bidegree (p, q) in $z, \bar{z}$, orthogonalised). Under the rotation $z \mapsto e^{i\theta} z$ every relevant object has *charge one*, so only the bidegrees $(1,0), (2,1), (3,2), \ldots$ (degrees $1, 3, 5, \ldots$) appear. The standard complex Gaussian measure, its rotation invariance and radial-moment reduction form a shared Lean foundation. The upper rounding proof depends on this foundation independently of the lower row certificates.

2.1 Upper bounds: rounding schemes

Given unit vectors, project a Gaussian onto them and round: $\varepsilon_i = f(\langle G, x_i\rangle)$ with $|f| \le 1$. The correlation of two rounded values depends only on $t = \langle x_i, y_j\rangle$,

$$\mathbb{E}\big[\overline{f(\cdot)}\, g(\cdot)\big] = H(t) = \sum_{k \ge 0} c_k\, t|t|^{2k}.$$

If $H(t) = c\,t + r(t)$ with $\|r\|_A = \sum_k |c_k - c\,\delta_{k0}| < c$, a tensor-power argument (Krivine's trick) gives $K_G^{\mathbb{C}} \le 1/(c - \|r\|_A)$. The shared Gaussian phase identity is proved from the Hermite basis and Mehler formula. Our sharp rounding is a smoothed two-coordinate phase rounding, $F_0 = \mathrm{phase}\big(z + \sqrt{1+a^2}\, v(|z|^2)\, w\big)$ with a twelve-term Laguerre profile v, multiplied by the cubic Taylor polynomial of a damped polynomial phase φ: $T = F_0 P_3(\varphi)$, $P_3(x) = 1 + ix - x^2/2 - ix^3/6$. The identity $|P_3(x)|^2 = 1 - x^4(3 - x^2)/36$ and the proved bound $\varphi^2 \le 99/100$ (hence $|\varphi| < 1$) give $|T| \le 1$, and a disk-valued rounding is as good as a unimodular one. The phase certificate checks 119,077 leaves in 512 cubes through six native range facts, with an analytic far field. Its kernel is $H(t) = \sum E[n,m]\, p(t)^n (-\tau(t))^m$ with an odd polynomial p of degree 41 and τ of degree 15 with zero linear coefficient (any odd preprocessors of Wiener norm ≤ 1 are realised by tensor powers). Compared with $c\,t$, $c = 7119483103461/10^{13}$, the residual is at most $\delta = 455212621809/(5 \cdot 10^{17}) \approx 9.10425 \cdot 10^{-7}$:

$$K_G^{\mathbb{C}} \le \frac{1}{c - \delta} = \frac{500000000000000000}{355973699960428191} < 1.404598148840.$$

The whole difficulty is to keep *every* coefficient c_k, $k \ge 1$, small at once: improving the cubic coefficient c_1 tends to spoil $c_2, c_3, \ldots$, and the infinite tail has to be bounded too.

The endpoint proof is *work in progress*. One completed supporting calculation, `ProfileFirstGaussianCell`, proves $\int_0^1 (\frac{1}{2} - s)^2 e^{-s}\, ds = \frac{5}{4} - \frac{13}{4}e^{-1}$ and encloses it in $[0.054391816192, 0.054391816194]$ using proved exponential bounds. The complete profile and response estimates remain to be proved.

The pinned LeanCert dependency supplies a further completed supporting calculation in `LeanCertPilot`:

$$1 \le \int_0^1 \frac{2\exp(-u^2/(1+u^2))}{(1+u^2)^2}\, du \le \frac{5}{4}.$$

An explicit 16-cell dyadic certificate checks the enclosure and reciprocal domains with `decide +kernel`. A separate identity binds the integrand to the actual affine Gaussian phase integral for $u > 0$. Both theorems are audited without new native facts; the production Gaussian error estimates for the target endpoint remain work in progress.

The actual Taylor sources also admit a proved finite auxiliary reduction in `ProductionFiniteLaplace`. With softening $\eta = 10^{-15}$ and cutoff $T = (2^{53})^2$, softening plus truncation costs less than $5 \cdot 10^{-15}$ in the full Gaussian L^2 norm. Smoothing and every Hermite-band projection preserve this bound. The regular auxiliary interval is $[0, 2^{53}]$, with $\eta^2 T \ge 40$. A LeanCert certificate proves $e^{-40} \le 10^{-17}$ by kernel reduction. The finite Fubini identity binds the truncated response to the coefficient integrals; their numerical quadrature remains work in progress. `ProductionCompactLaplace` also restricts the first squared Gaussian radius to $[0, 80]$. Its radial error is at most 10^{-17}, and the combined softening and both truncation errors still fit the same $5 \cdot 10^{-15}$ reserve. `ProfileAdaptiveSource` permits the shorter regular auxiliary endpoint 100 on any certified radial region with $s \ge 1/16$ and $|\kappa(s)| \le 1/100$. It retains 2^{53} elsewhere and proves that the combined source error for all selected regions still fits the original $5 \cdot 10^{-15}$ reserve, including after smoothing and band projection. The retained-cell certificates remain work in progress. The finite-union Gaussian pairing is identified with its rectangle integrals in `ProfileIntervalQuadrature`. Charge-one Parseval in `ProfileIntervalBands` then transfers retained coefficient-energy certificates to the original bands, without repeating the source reserve for each interval or coefficient. `ProductionRectangleAssembly` identifies those coefficients with a sum of retained nonoverlapping rectangles of the continuous unit-mode evaluator. An exact substitution $x = 1/u$ maps each auxiliary tail to a bounded interval with positive rational lower endpoint, including the Jacobian x^{-2}. No extra truncation or endpoint-zero premise is used. Finite partitions of these positive reciprocal intervals follow by mapping their boundaries back to the original coordinate. `ProductionMeshEnclosure` sums the certified regular and reciprocal cell radii, with the Gaussian normalization, into an enclosure of the actual retained coefficient. The common source reserve is applied afterward at the band level. The numerical cell and energy certificates remain open. The concrete partition in `SmallProfileMesh` uses seven radial strips. Kernel-reduced centered Horner checks establish the actual small-profile condition on three selected rational intervals. Thus its retained-source bound has no remaining profile-region premise; the numerical retained-energy certificates remain to be completed. Eight complementary centered-Horner rows in `SmallProfileSeparation` combine with the existing profile cover to give $|\kappa(s)| > 1/1000$ on the four other strips. Hence $\alpha x^2 + \kappa(s)^2 > 10^{-6}$ for $\alpha \ge 0$, uniformly even when the reciprocal coordinate is small.

The local affine-cell enclosure only requires continuity on the closed positive-coordinate rectangle. Coordinate clamping gives a continuous extension with the same integral. The complete normalized retained coefficients are real. Rational coefficient centers and radii give a finite sum of squared residual radii, which bounds the retained band by Parseval. The original source reserve is added once after this sum. The actual numerical enclosures and the endpoint remain work in progress.

The retained transfer is specialized to the fourteen actual captured bands. Their finite degree cutoffs and reserve inequalities are checked by kernel reduction. The source powers, stored centers and original caps are fixed; the remaining inputs are the numerical coefficient enclosures and their rational residual-energy checks.

Adaptive rectangular quadrature uses a binary tree of exact rational midpoint splits. Each leaf lies

in the original closed rectangle, and the original iterated integral equals the sum of the ordered leaf integrals. The auxiliary coordinate remains outside the radial coordinate; continuity is required only on the closed root. Actual numerical certificates for the complete leaf lists still have to be supplied.

Certified intervals for those leaves add to a root-integral interval by exact rational arithmetic. Pointwise analytic envelopes also give integral bounds by multiplying by the exact finite cell area. These operations preserve the separate source-error reserve and the original band budgets.

The fourteen regular and reciprocal roots are identified with the actual retained Gaussian coefficient, including the reciprocal Jacobian and local continuity. Rational bounds for $\sqrt{\pi}$ enclose its reciprocal, so interval multiplication accounts for normalization. Complete leaf certificates and rational endpoint checks then imply the coefficient bounds consumed by the original bands. The numerical leaf and energy certificates remain work in progress.

Exact midpoint identities also combine already certified subtree intervals. The normalized coefficient can be assembled from the fourteen resulting root intervals, permitting separate proof batches without flattening the complete leaf list.

A second route uses a finite-head estimate for the actual kernel directly; its coefficient enclosures and numerical head estimate remain work in progress.

Selected-spectrum comparison. The finite head only sees $S = \{(n, m) : n + 3m \le 501\}$. Write its approximation error as $G_T - X = \sum_r D_r$, where D_r is supported in degree band B_r with $\|D_r\|_2 \le e_r$. Let $w_p \in [0, 1]$ bound the retained absolute coefficient sum of the preprocessed block p. For nonnegative radii $A_r^2 \ge \sum_{p \in S \cap B_r} w_p^2 E_X(p)$, Lean proves

$$\sum_{p \in S} w_p |E_{G_T}(p) - E_X(p)| \le 2 \sum_r A_r e_r + \left(\sum_{g=0}^{3} \sqrt{\sum_{r \in I_g} e_r^2} \right)^2 .$$

Here I_0 contains the rounding remainder and I_1, I_2, I_3 are the three Taylor orders. Their bands are disjoint within each group, so their errors combine in quadrature before the groups are added. All pieces may share coordinates. Lean proves each coefficient cost by outward interval polynomial arithmetic; native arithmetic checks eighteen weighted radii from the stored table and its error allowance. The Gaussian component-error enclosures and their identification with the pieces remain explicit inputs.

Exact phase moments. For $\varphi = De^{-\varepsilon(U+V)} S(U, V, W)$, clear the rational coefficients as $A = LS$. Each even moment is a finite rational sum over the coefficients of A^{2j}, divided by L^{2j}, at Gaussian rate $1 + 2j\varepsilon$. The integral identity applies directly to the stored phase and does not use the numerical phase-cover certificate. Separate native checks give exact rational values for $\mathbb{E}\varphi^2$, $\mathbb{E}\varphi^4$, and $\mathbb{E}\varphi^6$, approximately $2.56852947444587 \cdot 10^{-6}$, $5.71439509703932 \cdot 10^{-10}$, and $1.68238736430120 \cdot 10^{-11}$. These equal the squared norms of the three positive Taylor sources. The same finite evaluator with a restricted Gaussian measure also proves $\int_{|z_1|^2 > 50} \varphi^2 \, d\gamma_2 < (5 \cdot 10^{-15})^2$. A native check bounds the exact rational factor after extracting e^{-155}; the Taylor polynomial through degree twelve at 5 gives $e^5 > 148$, hence $e^{-155} < 148^{-31}$. This certifies the radial tail of the production phase. Since $|\varphi| < 1$, every positive Taylor source restricted to this event also has $L^2(\gamma_2)$ norm below $5 \cdot 10^{-15}$. Sharp comparison-band errors still need the captured Hermite energy and approximation bounds; the total moments alone do not change the upper bound. The orthogonal tail interface subtracts the separate captured energies $\sum_i \max(0, \ell_i - e_i)^2$ from the total moment, retaining each band's error cap. An alternative interface uses one real inner product and the approximant norm to bound the whole approximation error. The remaining numerical coefficient and integral bounds are explicit hypotheses. Finite normalized Hermite expansions have exact rational squared norms. The preserved coefficient inputs supply a complete finite

check for the third approximant: all 6,552 coefficients through degree 51, its total norm, and its four projected band norms. Injective graded indexing also proves that its higher-degree tail vanishes. Its Gaussian approximation error remains a separate obligation; the stored error arrays are not treated as proofs. A continuous finite-polynomial representative satisfies the charge-one function interface. The third-order tail cap follows from four explicit finite-band Gaussian error caps by subtracting captured energy from the exact sixth moment; the Gaussian caps themselves remain hypotheses. The finite rounding difference $c_1 + c_2/2 + c_3/6 - v$ has norm at most 10^{-18}, proved by exact integer squares at a common dyadic denominator. The three other preserved center sources and their successful decoding are checked separately from Gaussian approximation errors. The first and second centers also have exact certificates for their six and four captured bands. The concrete conditional theorem fixes every center and its polynomial representative and derives the rounding and all three omitted tails internally. Its remaining inputs are fourteen explicit finite-band Gaussian error inequalities and the concrete energy-table accuracy. The slightly tighter finite error caps are generated from the original allowances; their Gaussian validity remains open. A Gaussian Laplace representation of complex phase is proved exactly, including the zero argument, integrability and a uniform absolute-integral bound. Absolute Fubini exchange applies this representation to the actual Taylor inner products. Affine completion of squares and translated Gaussian Wick sums give exact conditional monomial integrals with the inverse-rate mass. Finite support extends this evaluation to the same complex polynomials used by the Hermite basis, with integrability proved term by term. For each finite center, fixing the first coordinate gives its exact polynomial slice in the second coordinate. The actual source $\overline{c}(i\varphi)^j$ factors into this polynomial and Gaussian damping with conditional rate $\alpha = 1 + j\varepsilon$, giving an exact conditional finite Wick sum. Fixed-parameter boundedness justifies Gaussian product Fubini, and the actual inner product becomes the auxiliary and first-coordinate Gaussian integral of this finite evaluator, with no numerical integration premise. For each unit Hermite mode, a rational Wick polynomial gives the exact regular quadrature kernel in squared-radius form. A pointwise polynomial enclosure on a normalized square implies a bound for the real production cell integral and shows its imaginary part vanishes. Shared moment recurrences agree with the rational Wick compiler for positive inverse rate; finite table reuse requires proved base and successor equations. The real moments and their affine combination admit inverse-rate factorizations, and reciprocal-variable rate identities remain valid when the profile vanishes. These identities provide no numerical enclosure. Smooth angular moments

$$M_h(x) = \int_0^{\pi/2} \sin^{2h}\theta \, \cos^2\theta \, e^{-x\sin^2\theta} \, d\theta$$

are positive, have consecutive ratios in $(0, 1]$, and satisfy the exact three-term recurrence. For $x \geq 0$ their ratio update equals a monotone abstract update, for which finite backward interval propagation is proved from a terminal interval in $[0, 1]$. The angular ratio is bounded by the iterations with terminal values zero and one. The original moment integrand is interval integrable even at order zero, and the continuity, derivative, range and open-interval pullback identity for the angular substitution are proved. The integral change of variables gives $J_h(x) = (4/\pi)M_h(x)$, with endpoint equality handled almost everywhere. The original moments inherit positivity, the recurrence, and finite backward ratio enclosures for $x \geq 0$. Seed formulas and numerical node certificates remain open. The cell enclosure is an explicit hypothesis; no numerical cell certificate or improper-integral bound is claimed. The recurrence $M_{p+1,q} = \mu M_{p,q} + (q/\beta)M_{p,q-1}$ preserves the affine cancellation for every mixed degree, including zero mean and $q = 0$. Real radial powers with exponent greater than -2 also have their exact damped Gamma integrals and integrability proved. The zeroth conditional bracket simplifies exactly to $\alpha z/\beta^2$, where $\beta = \alpha + a^2 t$. For $|z| \geq r_0 > 0$, its conditional-kernel norm has the profile-uniform majorant $2t^{-5/2}/(\alpha r_0^3)$ and an integrated tail above $T > 0$ bounded by $4T^{-3/2}/(3\alpha r_0^3)$, including when $a = 0$. The full polynomial-weighted source tails, the radial endpoint and numerical integration enclosures remain open.

Radial repair has a quantitative ceiling. Lean certifies the polynomial disk map $z \mapsto zS_6(1-|z|^2)$, where $S_6(q) = \sum_{j=0}^{6} \binom{2j}{j} q^j/4^j$: the exact defect $1-(1-q)S_6(q)^2$ is q^7 times a polynomial with positive coefficients. Applied to $P_3(\varphi)$ this is a degree-39 factor. But Lean also proves $(1-|P_3(x)|)^2 \le x^6/1225$ for $x^2 \le 1$. For charge-one responses and any same feasible preprocessors, the full coefficient change obeys

$$\|K_H - K_G\|_A \le (\|H\|_2 + \|G\|_2)\|H-G\|_2.$$

Thus outward radial changes of disk responses cost at most $2\sqrt{\mathbb{E}\varphi^6/1225}$ in the full kernel. The integrated Lean theorem retains the sixth-moment and radial-gap premises explicitly; it does not identify the concrete Gaussian moment or certify a new score. These results leave the endpoints unchanged. Preprocessor changes must be compared at the same P, Q; this ceiling does not bound a separate gain from reoptimising them.

2.2 Lower bounds: Hermite projection games

A lower bound needs an operator T whose *vector* value is close to 1 while its *scalar* value is small. We use

$$T = P_{1,0} - \epsilon\, P_{2,1}, \qquad \epsilon = \tfrac{149}{200},$$

the projection onto linear functions minus a multiple of the projection onto the cubic chaos.

- *Vector value.* With $u_\ell = z_\ell/\|z\|$, $\sum_\ell \langle u_\ell, Tu_\ell\rangle \ge 1-(1+\epsilon)/(n+1) \to 1$.
- *Scalar value.* For all $|f|, |g| \le 1$, $|\langle f, Tg\rangle| \le B = 0.713420886479$.

A finite-partition transfer turns this into matrices, so $K_G^{\mathbb{C}} \ge 1/B > 1.401697117301$. The complete cover has 10,909 boxes. Two added pure-row law nodes and three mixed-source Young nodes use the original source caps and Stein cuts, so the Gaussian assumptions are unchanged. Their batches, every box and the full partition are checked separately. Adaptive binary splits reprice the existing components while preserving their scale interpolation domains. All generation inputs are retained in `data/lower-unshifted/`.

The scalar cap is the hard part. Writing $h = \frac{1}{2}(f+g)$, $k = \frac{1}{2}(f-g)$,

$$\Re\langle f, Tg\rangle = t^2 - \epsilon\|P_{2,1}h\|^2 - \|P_{1,0}k\|^2 + \epsilon\|P_{2,1}k\|^2, \qquad t = \|P_{1,0}h\|.$$

Only the last term is positive: we must show that a *bounded* function cannot have too large a cubic part without paying elsewhere. After a rotation $P_{1,0}h = t\, z_1$, and the cubic part of k splits into pieces that involve z_1 (*visible* sectors) and a *remote* residual $q \in H_{2,1}$ in the other coordinates. The proof then has three layers.

1. **Row reduction.** For each of 19 profiles r, the contribution of the visible sectors is the supremum of an explicit finite-dimensional Gaussian integral, a convex one-variable *potential* Ψ_r, evaluated at the scale of the remote residual. One gate variable is shared across rows.
2. **The law of the remote cubic.** What remains is a statement about the distribution of $U = |q|^2/\mathbb{E}|q|^2$ for an arbitrary cubic Wick polynomial in arbitrarily many variables. We use exact identities and inequalities valid for *every* such cubic: the sharp fourth-moment cap $\mathbb{E}U^2 \le 44$; the sixth-moment bound $\mathbb{E}U^3 \le 9m - 12 + 40.45313\,(m-2)^{3/2}$, $m = \mathbb{E}U^2$, which rests on thirteen tensor-contraction facts including the Wick expansion of the sixth cumulant; a first-moment bound; and families of *Stein cuts* $\mathbb{E}\,G(U) \ge 0$, obtained by Gaussian integration by parts and completing squares against spectral budgets, a covariance order and *contraction-Gram budgets* (positive semidefinite Gram matrices of the coefficient tensor).

3. **A finite certificate.** The parameters are covered by 10,762 boxes in (t, τ, x); on each box a positive combination of rows and node certificates (351 existing mixed nodes, two added pure-row law nodes, three added mixed-source Young nodes and 9,237 row nodes, each a polynomial majorant checked in exact rational arithmetic) bounds the scalar value by B.

2.3 The primal/dual viewpoint

For the real constant, Heilman, Jones and Malavolta proved an exact minimax: over rounding schemes μ with kernel coefficients $c_m(\mu)$ and Hermite projection games $Q = \sum_m q_m \Pi_m$,

$$\frac{1}{K_G} = \sup_{\mu} \Big[c_1(\mu) - \sum_{m \geq 3} |c_m(\mu)| \Big] = \inf_{Q} \ \mathrm{OPT}(Q),$$

The complex programme uses charge-one bidegrees. A global complex minimax identity is not established here; the two endpoint proofs use separate one-sided transfer theorems. An upper bound comes from a realizable kernel, and a lower bound from a game with a uniform scalar norm cap. Finite-pool duality alone does not certify either global claim.

3 What exactly the computer checks

Proved in Lean	**Assumed (certified outside Lean)**
Lower bound. `lower_bound_of_refined_cover_data : NodeCaps → IsGrothendieckBound K → 1.401697117301 < K` the witness T on L^2 of the Gaussian measure and its vector value; the transfer to matrices; the polarisation identity; the per-row Gaussian reduction of $\langle f, Tg\rangle$ for every unimodular pair (frames, fibre decomposition, Parseval, gates); all moment, Stein, spectral and contraction-Gram inequalities for every circular cubic, including the sixth-cumulant Wick identity; the finite certificate (five row, three original-batch, one added law-batch and five refined-cover native checks)	`NodeCaps`: the 19 potentials Ψ_r (defined in Lean) lie below their stored rational caps at the tabulated nodes. `lower_bound_of_capsRest` proves the same endpoint from `CapsRest`: the vertex caps of rows 12–15 ($t = 0.84$–0.855) and any excluded-node caps, using the same refined cover. The 60 range checks for fifteen rows use a computation-only layer. CI replays the final assembly when the 53-file cache provenance matches; a mismatch triggers the runner build and the axiom audit. Four rows retain external vertex caps.
Upper bound. `UpperChaos.fresh_upper_rounder`, for an L^2 charge-one truncation X under component-error and table-accuracy inputs, proves $K_G^{\mathbb{C}} \le 500000000000000000/355973699960428191$. the concrete disk-valued rounding; tensor Hermite completeness by Fubini, Gaussian mixing by characteristic functions, Parseval and Mehler, charge selection, summability; the transfer inequality (tensor powers); kernel composition and locality; the head through degree 501 from 21,168 stored energies by a verified interval evaluator (one `native_decide`); the tail bound; the endpoint; the selected-block comparison and native band masses	`ComparisonInputs`: an exact error decomposition, spectral supports and inherited Gaussian component norm caps. The concrete Taylor specialization leaves fourteen finite Gaussian band-error estimates and the energy-table discrepancy as hypotheses. Lean derives the selected comparison $851922301/10^{15}$ (93.57% of the residual allowance). The decomposition itself is explicit in Lean (`difference_pieces`, `fresh_upper_pieces`): 18 band pieces of the smoothed Taylor errors and the rounding remainder, so only their 18 norm bounds remain external. Finite table accuracy $6847009/(5 \cdot 10^{17})$. The Hardy tail $303362486/10^{16}$ (3.33%, weighted square sum $\le 1737972/1000000$ at radius $4149/4000$) is proved (`HardyReal.hardy_GT_tight`). The native head is $28153/10^{12}$ (3.09%).

The repository-wide audit also covers the native facts pinned in `data/axioms/native.txt`, read by `scripts/check_axioms.py`. These check the upper head, selected-band masses, the six-part phase cover, the second, fourth and sixth phase moments and the squared radial phase tail, cap chords, the lower row/batch certificates, the additional pure-row and mixed-source law nodes and the complete finite lower cover. The native facts (cap-range checks, Hardy region and circle checks, and others) are listed in `data/axioms/native.txt`; the report is `scratch/Axioms.lean`, with its required declarations in `data/axioms/required.txt`. No `sorry`; normal `ready/*` delivery CI builds the single target `Grothe`, cap rows included, and checks its one axiom report before merging. The cap-range checks take about 21 CPU-hours cold, so CI replays them from a verified build cache. Its provenance hash covers 48 local computation modules and five directly hashed extras (53 files total); the extras' imports are not traversed. The numerical checker has its own precompiled library, `GrotheCapsKernel`, whose constants retain the `GrotheKernel` namespace. The caps digest is part of the build-cache key: when no verified cache matches, CI runs the same check on the self-hosted runner, which builds, audits and marks `.lake/build/caps-closure.sha` before saving the cache. That persistent build supplies later CI replay; no separate manual droplet

verification is needed. A digest alone is not verification. The localised-piece checkers likewise import only computable definitions and rational data; their campaign and final wiring remain separate. The generic response and quintic-transfer lemmas add no native facts or endpoint assumptions. The lower bound still needs numerical caps; the sharp upper bound retains external Gaussian component-error and table-accuracy inputs. The four existing arc checks each give at most 273/100; the proved bound on π turns their sum into the circle-mean budget 1737972/1000000. Before the tail Cauchy–Schwarz step, the known linear coefficient's weighted square is deducted from that budget. The abstract table interface uses this sharper Hardy input; the concrete rounding proves it without new Gaussian hypotheses. Holomorphy under positive kernel decay, odd-series continuation and circle Parseval are proved. The holomorphy and Hardy assembly also cover separate measurable disk-valued left and right responses (`HholPair`, `hardy_of_pair_inputs`); the incumbent results are specializations. The decay region and the circle mean are checked in Lean by five interval facts. With them, `hardy_of_real` reduces `hardy` to summability and the real Gaussian identification of the odd series, which `HardyReal.hreal_GT` proves for G_T: the smoothed Mehler pair has the law of $(U, RU + SW)$ (characteristic functions), whose density on $\mathbb{R}^8$ is the holomorphic kernel at the real cross covariance (Cholesky substitution). Hence `hardy_GT`: the Hardy input is proved. The research modules also prove 13 algebraic facts for coupled complex modulus majorizers, finite descent and semiconvex chord assembly, using only the three standard axioms. Their Gaussian integration, response extraction and global-cover obligations remain separate; these facts change neither endpoint nor its hypotheses.

4 Discoveries along the way

Status labels: [**Lean**] machine-checked in this repository; [**P**] written proof; [**P,C**] proof with an exact or interval-arithmetic certificate replayed independently; [**F**] floating-point evidence only.

Other upper-bound variants and their verification records are preserved outside the active build in `miscellaneous/other-bounds/`. Shared Gaussian phase identities and Laplace representations remain part of the sharp target.

4.1 Common responses, common tails, and an architecture separator

[**P; supporting algebra in Lean**] A symmetry-closed family of actual cubic responses has an exact five-mode first/cubic image and a fixed-size original-noise replica representation. It preserves the shared quadratic drivers: a scalar Gaussian used to describe a cubic's law must retain third-chaos noise correlation ρ^3, rather than acquire first-chaos correlation ρ.

[**Lean**] All correlation samples of a response pair use the same signed Hermite coefficients. Prices a_i with $|\sum_i a_i \rho_i^{2k+1}| \le 1$ for all $k \ge 2$ bound the complete omitted absolute tail. A finite front and one geometric remainder budget certify every index. The resulting separator is linear in response columns, so it bounds whole convex pools, with explicit integration-error costs. Numerical pool screens still require certified input integrals; a favorable relaxation value is not an achievable rounding score.

[**Lean**] An exchange can retain the incumbent's signed coefficients with one error allowance. If $\sum_k |c_k - \widehat{c}_k| \le \xi$, then every admissible spectral price has error at most ξ, and the full reciprocal score satisfies $D(c) \ge D(\widehat{c}) - \xi$. The same coefficient error must not be charged independently at every sampled correlation. The current bounded phase-perturbation searches use this interface for discovery; their Gaussian integrals still require certification before they can improve either endpoint.

[**P,C; polynomial certificate in Lean**] The entire Gaussian-phase cubic-preprocessor family $H_{\mathrm{phys}}(A\rho + B\rho^3)$, $|A| + |B| \le 1$, including arbitrary complex phases and convex mixtures, has full reciprocal score at most 0.71186. This is below the incumbent $0.7119473997\ldots$. The separator is

$\Re(c_0 - \frac{9}{25}c_1 - c_2 - c_3)$; positive polynomial certificates handle all complex phases. Thus optimizing this class further cannot improve the current upper bound. General non-Gaussian cubic responses and nonlinear shared image fields are not excluded. The Gaussian-family coefficient extraction is a written application of the Gaussian phase identity; it is not yet connected end to end in Lean.

Phase errors and exact symmetries [Lean]. Phase alignment has an exact modulus-weighted quadratic defect. An unweighted degree-moment budget is unavailable: the phase coefficients satisfy $r_k \geq 1/(4k(k+1))$ for $k \geq 1$, so $\sum_k (2k+1)c_k = \infty$. Certificate comparison can instead align an anchor by a common isometry: if the measured blocks are transported with it and the remaining error is at most e, the complete energy-spectrum discrepancy is at most $2\|x\|e + e^2$. A pure symmetry has zero price, including its entire tail. The kernel is preserved only when its correlation operator is transported too; rotating only the response can change the anisotropic kernel used here. An additional polarized certificate prices the whole omitted energy change by AB, where A, B bound the approximation errors of the response difference and sum by retained polynomials. These may be certified with positive squared-error integrals, avoiding subtraction of nearly equal moments in a residual Gram matrix. Two overlapping coordinate tests either keep the phase argument away from zero or resolve it by an affine polar chart. Lean checks the exact transformed Gaussian density and a Bernstein range enclosure. A symmetry-invariant cutoff preserves admissibility and costs at most its omitted mass in squared L^2 distance. This lets an upper construction use compact chart domains with an explicit full-spectrum allowance. An exact rational phase update has unit modulus for every real field, with a cancellation-free increment formula. A finite radius jet retains the radius and angle as independent polynomial variables until cancellations finish. Lean checks carry-free coefficient indexing, exact translation, and the finite remainder bound on complex patches. A norm-only homogeneous transport certificate also has a proved ceiling: its certified score cannot exceed b/N^2 when the anchor has norm $N > 0$ and score at least $0 \leq b \leq N^2$. This limits that certificate formula, not the corrected response's true score. The update assembly keeps the signed retained changes, passes the positive error price through contractive smoothing and the bivariate kernel, and connects the entire score to the transfer theorem. The reference error is the inherited anchor error plus finite weighted retained-data errors, charged once. These forty-seven Lean results change neither endpoint. The compact Gaussian integrals still require certification.

4.2 Inequalities for cubic Gaussian chaos

Throughout, q is a circular cubic Wick polynomial (bidegree $(2,1)$) in any number of complex Gaussian variables, normalised by $\mathbb{E}|q|^2 = 1$, and $U = |q|^2$.

- **Sharp fourth moment [Lean].** Every real third-chaos variable satisfies $\mathbb{E}F^4 \leq 93\,(\mathbb{E}F^2)^2$, with equality for $x^3 - 3x$. For the cubic kernel, write $N = \sum f^2$, $s = \|f \star f\|^2$ and $h = \|\operatorname{Sym}(f \star f)\|^2$. The Wick identity $\mathbb{E}F^4 - 3(\mathbb{E}F^2)^2 = 1296(s + 3h/2)$, together with $\mathbb{E}F^2 = 6N$ and $h \leq s \leq N^2$, proves the bound directly, without a separate positivity premise on the Gaussian expectation. For a circular cubic, the complex contraction identities and Hilbert–Schmidt estimates instead give $\mathbb{E}U^2 \leq 44$ at any dimension. Equality is attained by $Z(|Z|^2 - 2)/\sqrt{2}$ for a standard complex Gaussian Z; the equality calculation is analytic. The Lean theorem is `Circular.fourth_moment_le_variance`. It implies $x \leq \sqrt{42}$, while the existing certificate still uses $x \leq 31/4$.
- **Sixth moment [Lean].** $\mathbb{E}U^3 \leq 9m - 12 + C_6(m-2)^{3/2}$ with $m = \mathbb{E}U^2$ and $C_6 = 40.45313$. The Wick monomial mixed with Gaussian noise attains 40.295, so the constant is sharp to 0.4%. The inequality is *false* without circularity: two real cubics $He_3(x_0)/\sqrt{12}$, $He_3(x_1)/\sqrt{12}$ give $m = 47$ and $\mathbb{E}U^3 = 13506 > 12622.55$.
- **The sixth cumulant as a sum over graphs [Lean].** The Wick identity for κ_6 of a cubic

chaos is proved by expanding $\mathbb{E}F^2, \mathbb{E}F^4, \mathbb{E}F^6$ over closed cubic multigraphs, one Gaussian integration by parts per edge ($69 + 15 + 3$ certified steps). The disconnected graphs cancel, and the six connected cubic multigraphs on six vertices give the tensor contractions.

- **Contraction-Gram budgets [Lean].** A new family of universal inequalities between the chaos components of U and of q^2. With $X = \Gamma(q, \bar{q})/3 - 1$, $c = \Gamma(q, q)/3$, $r = \Re(c\bar{q}^2)/U$, $P = \mathbb{E}[XU]$, $Q = \mathbb{E}[rU]$, for example

$$\mathbb{E}X^2 + \tfrac{1}{10}\mathbb{E}r^2 \le \tfrac{34}{45}P - \tfrac{1}{18}Q.$$

  Each budget is certified by two positive semidefinite 2×2 Gram matrices of contractions of the coefficient tensor.

- **First moment [P,C].** $\mathbb{E}|q| \le 0.97073$, from a two-plane small-ball estimate (a winding argument on S^3 and a horizontal Lipschitz bound). The sharp value $\sqrt{\pi}/2 = 0.8862$, which the Gaussian attains, is open.

A common budget for the same tensor. The two joint sixth-moment bounds share one tensor and scalar argument in `JointSixthCore.lean`: Hilbert–Schmidt and cubic-trace bounds, a completed square, and Young's inequality. The production cuts and parameterized refinements specialize that argument through the same real-to-complex normalization identities. The sixth moment and the Stein fluctuations cannot be optimized independently: both depend on the same contraction invariants. We now formalize this connection in Lean, including the real/complex normalization and a finite checker that reserves part of the contraction budget before accepting a Stein cut. Signed reserve coefficients are retained. Ordinary and joint Gram budgets specialize one signed-reserve soundness theorem. Likewise, ordinary spectral cuts specialize the general shifted-cut theorem, and Stein and spectral nodes use the same polynomial-majorant integration lemma for an arbitrary integrable cut with nonnegative expectation. The endpoint uses 41 such nodes, with each node restricted to its proved kurtosis range; the full checker and analytic bridge are in Lean. Retaining the upper bound on the negative cumulant contraction yields a stronger parameterized moment family. An exact transport identity then upgrades a node's majorant when its budget permits a larger moment-cut weight, without a new half-line search. These two refinements are also formalized in Lean. The moment costs also admit a convex formulation with two 2×2 PSD blocks and linear constraints. An exact rational checker connects each accepted recipe to the actual cubic moment inequality, replacing the outer parameter grid. Its exact scalar boundary is also proved in Lean. For moment weight $w > 0$ and s_0 cost $A > 0$, the minimum h_0 cost is

$$B_{\min}(w, A) = \begin{cases} 11781w/32 + 18225w^2/(4A), & 11A \le 360w, \\ 198w + 22500w^2/(A + 40w), & 11A \ge 360w. \end{cases}$$

Both branches have explicit feasible recipes. A three-rational checker for (w, A, B) therefore replaces the auxiliary moment-cost variables, and Lean connects its acceptance to the actual cubic on the checked node range. The contraction reserve and pointwise majorant still need their own exact checks. This is optimality within the certificate family; it does not identify all attainable response laws.

4.3 The remote functional

Let $L = z_1(a + b|z_1|^2)$ with the constants of the witness, and for a cubic G put $\Lambda(G) = \mathbb{E}|L + G| + \mathbb{E}|L - G| - \mathbb{E}|G|^2/\epsilon$ and $S = \sup_v\{2\mathbb{E}|L + \sqrt{v}U| - v/\epsilon\}$, U a complex Gaussian. The conjecture $\Lambda(G) \le S$ for all cubics ("Conjecture K") is the analytic heart of the lower bound.

- **A one-function reformulation [P].** Circular symmetry, $\mathbb{E}|W| = (\pi/2)\mathbb{E}|\Re W|$, and the identity $|a+b| + |a-b| = 2\max(|a|, |b|)$ for real a, b give exactly

$$\Lambda(G) = 2\mathbb{E}|L| + \pi\,\mathbb{E}\big(|\Re G| - |\Re L|\big)_+ - \mathbb{E}|G|^2/\epsilon,$$

  so $\sup_G \Lambda - 2\mathbb{E}|L| = \sup_t \big[\frac{\pi^2\epsilon}{4}\|P_{2,1}t\|^2 - \pi\mathbb{E}(|t|\,|\Re L|)\big]$ over real functions $|t| \le 1$. All of the gain sits near the hyperplane $\Re z_1 = 0$.

- **The Gaussian is a strict local maximum [P,C].** At the Gaussian point every direction that involves z_1 has negative second variation, certified in interval arithmetic for seven sector coefficients. The Hessian's kernel is the entire z_1-independent cubic sector; a negative fourth-order term controls those genuine cubic directions.

- **The independent sector [P,C].** For every cubic q not involving z_1, $\Lambda(q) < S + 3.6 \cdot 10^{-7}$. Moment inequalities alone stall about 10^{-5} above S, because $h(x) = 2\mathbb{E}|L + \sqrt{x}|$ is a Bernstein function and the three-moment relaxation is maximised by a two-point law [P]. Going below that floor needs the Stein and spectral cuts.

- **Two natural shortcuts are false [P,C].** $|q|^2$ does not dominate $\mathbb{E}|q|^2 \cdot \mathrm{Exp}(1)$ in convex order, and there is no Gaussian Laplace domination: $Q = X(4(|X|^2 - 2) + 3i(|Y|^2 - 1))/\sqrt{41}$ has $\mathbb{E}e^{-20|Q|^2} < 0.0453 < 1/21$.

4.4 Actual responses

Instead of bounding Λ for all cubics, one can use that the cubic comes from a bounded response. Write F for the cubic part of the response and $\mathcal{N}$ for the value of the witness on it.

- **Exact support [P].** The paired support $\frac{1}{2}(\mathbb{E}|A+B| + \mathbb{E}|A-B|)$ is exactly the pointwise support function of the complex response set $\{(h,k) = (e^{i\alpha}\cos\theta,\ ie^{i\alpha}\sin\theta)\}$, because $(|A+B| + |A-B|)^2 = 4\max_\theta |A\cos\theta - iB\sin\theta|^2$.

- **Concentrated cubics are harmless [P,C].** At $\epsilon = 1891/2500$ on the central rows $0.84 \le t \le 0.851$, $\mathcal{N} < 0.71325 < 1/1.402$ whenever the cubic part of the response concentrates enough energy on a structured subspace:

  - 80% on one diagonal Wick direction;
  - 90% on one mixed monomial $X^2\bar{Y}$, or on a whole slice $\{X^2\bar{A}\}$;
  - 93.5%, 96.2%, 97.6%, 99.2% on any cubic in $2, 3, 4, 6$ remote coordinates;
  - 95.5% to 99.95% on $H_{2,0}(A)H_{0,1}(B)$ with $\dim A \le 24$, via a sharp support theorem whose extremiser has law $R_m G/\sqrt{m(m+1)}$, $R_m \sim \Gamma(m, 1)$.

  The remainder of the cubic may be arbitrary and fully coupled.

- **The obstruction to covering [P].** $Q_N = X_1(\sum_{j\le N}|X_j|^2 - N - 1)/\sqrt{N+1}$ keeps an influential direction while its projection onto every fixed-dimensional core tends to zero. So "small core or Gaussian" is not an exhaustive dichotomy.

- **How far from the benchmark structured cubics sit [F].** Remote-only payoffs relative to S:

sum of N slices, $N =$	1	2	4	8	16	24	∞ (Gaussian)
$S - \sup\Lambda$ (10^{-3})	1.68	1.42	1.08	0.73	0.43	0.32	0

  The tested concentrated cubics leave room of order 10^{-3}; these diffuse sums approach the benchmark. This evidence does not classify all near-maximizers.

4.5 Limits of the witness method

- **Ceilings [P,C] and [Lean].** Actual finite Gaussian responses give the scalar cubic family a ratio ceiling < 1.4022581233 when taking suprema over Gaussian dimensions, for every complex multiplier; its generic cancellation arithmetic is in Lean, its Gaussian construction and integral bounds are external. At $\epsilon = 149/200$ a genuine response gives norm > 0.71329, excluding 1.402 for that witness. Explicit "chirp" responses prove that games of degree ≤ 5 cannot exceed 1.4036705, and of degree ≤ 7 cannot exceed 1.4044895 [P,C].
- **Extending the pattern fails [P,C].** The full alternating game $\sum_{k\geq 0}(-1)^k P_{k+1,k}$ has $\sup_n \|\cdot\|_{\infty\to 1} = 1$, attained by explicit unimodular responses (one in complex dimension 308,835 is replayed). So better high-degree games must be re-optimised against new responses, not extrapolated.
- **Certified finite exchange [P].** For a finite response pool truncated at level D with value v_D, a global witness norm cap $v_D + \delta$ and a realizable mixture with omitted coefficient tail at most $\tau < v_D$ give $1/(v_D + \delta) \leq K_G^{\mathbb{C}} \leq 1/(v_D - \tau)$ by the two one-sided transfers. The reciprocal interval has width $\delta + \tau$; both the uniform witness cap and the complete mixture tail must actually be certified.
- **A global spherical form, and why it does not certify cheaply [P] and [Lean].** Finite Gaussian Hermite games transfer, in every Gaussian dimension, to compact operators on the sphere with the same bounded responses and the same shared drivers; the numerical bounds are unchanged. The dual operator can then be truncated with a rigorous error: the cubic multiplier tail and the signed cap assembly are proved in Lean (`SphericalTransfer.cubic_tail_32_12`, `global_cap_from_truncation`); the analytic bridge is a written proof (`docs/SPHERICAL_TRANSFER.md`). But the operator has no square-integrable kernel: its multiplicity-weighted squared coefficients grow like r^7. So no Hilbert–Schmidt-type certificate can control it, and direct coefficient optimisation is impractical (about $8 \cdot 10^{20}$ coefficients in the tested truncation). A useful norm cap must bound the signed action on bounded responses, exploiting the unitary symmetry.

5 Why the gap is hard

The witness class caps the lower bound. A game supported on degrees $\{1, 3\}$ sees only the cubic constraint on roundings. Even a perfect evaluation of the norm of $P_{1,0} - \epsilon P_{2,1}$ cannot certify more than 1.4022581233 (the certified family ceiling above). Proved obstructions are known for higher degrees: every game of degree ≤ 5 is capped at 1.4036705, and every game of degree ≤ 7 at 1.4044895, both below Haagerup's value. Reaching the upper end of the interval therefore needs games of degree ≥ 9; in the real case the certified game uses all odd degrees up to 21.

Certifying higher degree is the real bottleneck. Two different certificate architectures, gated rows and support functions, both lose about $2 \cdot 10^{-2}$ on the best degree-5 game. The reason is structural: a bounded function with a large cubic part is forced to have a degree-five part too. Through the visible coordinate, the cubic part sees the average of the remote coefficient profile and the quintic part its variation, and boundedness makes the profile vary. A certificate that treats the chaos degrees as independent cannot see this coupling. The coupling is *driver sharing*: in the dimension-limit model the best degree-5 game has norm 0.712464 ($1/N = 1.40358$) when its degree-5 mixed sectors $(|z_1|^2 - 1)W$ and $z_1^2\overline{W}$ are driven by the same remote cubic W as the cubic sector, and 0.724753 when they are driven by independent copies; the optimum then switches the degree-5 part off. So a higher-degree certificate must let one remote residual feed several degrees through the visible coordinate. The cubic rows already do this for their degree-3 sectors; extending them to the degree-5 companions of the same residual is the next construction [F].

Why the complex case is harder than the real one. For real ± 1 responses, $h = \frac{1}{2}(f+g)$ and $k = \frac{1}{2}(f-g)$ have disjoint supports, so the two sides of a game decouple at every degree. For complex unimodular responses $|h|^2 + |k|^2 = 1$ and $\Re(h\bar{k}) = 0$ force $f = e^{i\alpha}e^{i\theta}$, $g = e^{i\alpha}e^{-i\theta}$: one shared phase field α drives both sides, and they never separate. The same shared phase is why marginal laws do not determine the relevant nonlinear correlations: $(H(X), X)$ and $(H(Y), X)$, with $H(X) = X(|X|^2 - 2)/\sqrt{2}$ and X, Y independent, have identical marginals and covariances, but $\mathbb{E}[\bar{r}\, q|q|^2]$ equals $8\sqrt{2}$ in the first case and 0 in the second.

6 Open problems

Planned work is tracked as GitHub issues of the repository: epics #12 (unconditional lower bound), #17 (upper inputs), #22 (better lower bound), #29 (build infrastructure) and #34 (research plans).

1. Complete and verify the node caps (`CapsRest`, the remaining vertex-cap hypothesis), which would make the lower bound fully machine-checked. The caps of rows 12–15 are localised caps: coverage by constrained pieces is checked in Lean, and most pieces pass; the remaining pieces fail the exact positivity check of their S-lemma matrix by about 10^{-5} and need re-optimised multipliers.
2. Discharge the sharp upper rounding's numerical comparison and energy-table error; its Hermite expansion, structural analytic bridge and Hardy tail bound are formalized.
3. Certify a game of degree ≥ 5 with a *multi-grade fibre*: one bounded function of the visible coordinate and one remote direction per degree, so that the cross-degree coupling is kept. This is the route to a lower bound above 1.403.
4. Run a complex primal/dual search, alternating linear programs over games of degree $\leq D$ and pools of roundings, to locate $K_G^{\mathbb{C}}$ to about 10^{-4} and to test whether Haagerup's value $1/\gamma^{\dagger}$ is the answer.
5. Prove the sharp first-moment bound $\mathbb{E}|q| \leq \frac{\sqrt{\pi}}{2}\|q\|_2$ for cubic Wick polynomials. It is natural, it appears to be open, and it is the largest single lever on the law side.

How this was made

The constructions and certificates were found by several AI research systems working on a shared manuscript, cross-checking each other's packages by independent replay. The endpoint theorems have the explicit Lean hypothesis lists in Section 3, and their axioms are audited in CI. The discoveries labelled [P] and [C] also include paper proofs and external exact or interval certificates; they are not all formalized in Lean. Selected written proofs and reproducible certificate records accompany the repository; the larger working manuscript is maintained separately.

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